

Exp No: 9 ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

Program:

```
const int trig = 10;
```

```
const int echo = 9;
```

```
long duration;
```

```
float distance;
```

```
void setup()
```

```
{
```

```
  pinMode(trig, OUTPUT);
```

```
  pinMode(echo, INPUT);
```

```
  digitalWrite(trig, LOW);
```

```
  Serial.begin(9600);
```

```
}
```

```
void loop()
```

```
{
```

```
  digitalWrite(trig, LOW);
```

```
  delayMicroseconds(2);
```

```
  digitalWrite(trig, HIGH);
```

```
  delayMicroseconds(10);
```

```
  digitalWrite(trig, LOW);
```

```
duration = pulseIn(echo, HIGH);
```

```
distance = duration * 0.0343 / 2;
```

```
Serial.print("Distance = ");
```

```
Serial.print(distance);
```

```
Serial.println(" cm");
```

```
delay(500);
```

```
}
```

Output

